

Isabel Panicker

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Education

Georgia Institute of Technology, School of Chemical and Biomolecular Engineering, Atlanta, GA August 2023 - Current

PhD Candidate, Chemical and Biomolecular Engineering, Minor in Materials Science and Engineering

Focus: electron microscopy, single-particle tracking, nanotechnology, soft matter, materials' dynamics characterization

Advisor: Dr. Vida Jamali

Committee: Dr. Saad Bhamla (ChBE), Dr. David Flaherty (ChBE), Dr. Joshua Kacher (MSE), Dr. Hang Lu (ChBE)

Purdue University, Davidson School of Chemical Engineering, West Lafayette, IN

August 2019 - May 2023

B.S. Chemical Engineering, Minor in Management

Research Experience

PhD Candidate - Jamali Lab | Georgia Institute of Technology August 2023 - Current

- *Thesis:* Developing liquid-phase transmission electron microscopy (LPTEM) as a robust single-particle tracking tool to probe nanoscale transport and complex mechanics of soft matter systems.
- Applied canonical statistical diffusion analysis and deep-learning-based diffusion classification on LPTEM obtained single-particle trajectories of ~75nm gold nanorods to showcase how ionic charge environment modulates liquid cell interfacial chemistry which generates non-Brownian nanoparticle dynamics and establish the first passive nanorheology framework in LPTEM to model these dynamics, enabling quantitative extraction of complex viscoelastic moduli from both translation and rotational nanoparticle trajectories.
- Built SAM-EM, a custom Python-based tool to enhance particle tracking precision and acquisition throughput with LPTEM, finetuning Meta's Segment Anything Model 2 (SAM-2) on 94,000 simulated LPTEM frames, and integrated with particle tracking and statistical analysis tools, allowing for zero-shot segmentation of noisy LPTEM acquisitions.
- Revealed superdiffusion and resulting compositional polarity dependency of active ~100nm catalytic platinum-polystyrene Janus nanomotors with optical microscopy and LPTEM, demonstrating how catalytic asymmetry can continue to bias displacement direction despite thermal fluctuations unique to the nanoscale.
- Mapping nanoscale viscoelasticity of liquid-phase separated polymer systems using LPTEM-based passive nanorheology, revealing how spatially heterogeneous mechanics evolve at the nanoscale (~200nm).
- Visualized and characterized structural evolution of polymer-based Self-Agglomerating Nanohydrogels (SANGs) under variable environmental conditions as a mode of drug delivery.
- Mentored 10 undergraduate researchers (past and present) in laboratory skills, nanomaterial synthesis, and microscopy, facilitating independent experiments and meaningful contribution to research projects.

Undergraduate Researcher - Solar Energy Group | Purdue University

January 2022 - May 2023

- Developed and optimized solution-based synthesis protocols for multinary sulfide thin films (e.g., Cu₂BaSnS₄, alkaline earth metal chalcogenide perovskites) as energy efficient (< 500°F) alternatives for photovoltaic applications.
- Synthesized crystalline sulfide materials via low-temperature decomposition of thiol-metal precursors, screening thiol-metal combinations across solvent environments, confirming phase and composition using Raman spectroscopy, XRD, and XRF.
- Completed a senior thesis titled "Solution Based Synthesis of Alkaline Earth Metal Thiolate Precursors for Photovoltaically Active Multinary Sulfides".

Professional Experience

Drive Unit Quality Engineering Intern | Tesla Gigafactory, Reno, Nevada

May 2022 - August 2022

- Utilized SolidWorks to create a 3D printed tool to aid rundown in manufacturing process, increasing throughput.
- Established manufacturing instructions on individual Drive Unit part storage prior to workstation introduction.
- Managed acquisition and installation of a water in oil sensor inline for constant monitoring of water concentration.
- Designing and conducting cleanliness test to investigate part contamination resulting from current storage practices.
- Executed validation trials to ensure supplier compliance to dimension specs and prevent part starvation.

Selected Publications

Peer-Reviewed Articles

- **Panicker, I.**, Shabeeb, Z., Hargus, C., Jamali, V. "Solution-Tunable Interfacial Interaction Landscape Governs Anomalous Nanoparticle Diffusion in Liquid-Phase Electron Microscopy," *ACS Nano* (2026) (Featured on Front Cover).
- Shabeeb, Z., Sreeram, S., **Panicker, I.**, Wang, A., Xu, M., Goel, R., Jamali, V. "SAM-EM: Segment Anything for Liquid-Phase Electron Microscopy Video Segmentation," *Digital Discovery* (2026).
- Housley, S.N., Bourque, A.R., Matyunina, L.V., **Panicker, I.**, Echeverri, E.S., Izykiewicz, M., Herrmann, O.A., Lee, S., Arboleda, J.C., Jamali, V., McDonald, J.F., Dahlman, J.E., Finn, M.G. "Tumor Agnostic Drug Delivery with Self-Agglomerating Nanohydrogels (SANGs)," *Nature Communications* (2026).

- Agarwal, S., Vincent, K. C., Turnley, J. W., Hayes, D. C., Uible, M. C., Durán, I., Canizales, A. S. M., Khandelwal, S., **Panicker, I.**, Andoh, Z., Spilker, R. M., Ma, Q., Huang, L., Hwang, S., Kisslinger, K., Svatek, S., Antolin, E., Bart, S. C., & Agrawal, R. “Breaking Barriers in Chalcogenide Perovskite Synthesis: A Generalized Framework for Fabrication of BaMS₃ (M=Ti, Zr, Hf) Materials,” *Advanced Functional Materials* (2024).
- Turnley, J. W., Vincent, K. C., Pradhan, A. A., **Panicker, I.**, Swope, R., Uible, M. C., Bart, S. C., & Agrawal, R. (2022). “Solution Deposition for Chalcogenide Perovskites: A Low-Temperature Route to BaMS₃ Materials (M = Ti, Zr, Hf),” *Journal of the American Chemical Society* (2022).

Manuscripts Under Review and Preprints

- **Panicker, I.**, Roque, K.I., Ruiz, R., Ben-Moshe, A., Bustamante, C., Alivisatos, A.P., Jamali, V. “Polarity-resolved Imaging Reveals Active Transport in Catalytic Janus Nanomotors,” *ChemRxiv* (2026).

Selected Presentations and Panels

- “Revealing Superdiffusive Active Motion of Nanoscale Particles with Correlative Light and Electron Microscopy” - ACS Spring 2026.
- “Programming and Probing Interfacial Transport in Liquid-Phase TEM through Ion Screening and Passive Nanorheology” - ACS Spring 2026.
- “Programming and Probing Interfacial Transport in Liquid-Phase TEM through Ion Screening and Passive Nanorheology” - Invited Talk at 2026 Gordon Research Seminar on Liquid-Phase TEM (Spring 2026).
- “Screening Nanoparticle Surface Interactions at the Liquid-Solid Interface in Liquid-Phase Transmission Electron Microscopy” - Poster Presentation at Soft Matter Day (Spring 2025).
- “Developing a Single Particle Tracking Method through Correlative Microscopy” - Poster Presentation at American Chemical Society Colloid and Surface Science Symposium (Summer 2024).
- “Organizing a Successful Research Symposium” - Panelist at AIChE Annual Student Conference (Fall 2023).
- “Synthesizing Metal Thiolates as Soluble Precursors for Multinary Sulfides” - Poster Presentation (AIChE Fall 2023), (Purdue University AIChE URS Spring 2023).
- “Soluble Precursors for Chalcogenide Semiconductors Containing Alkaline Earth Metals” - Poster Presentation (Purdue University URC Spring 2022).

Honors and Awards

Georgia Tech ChBE Eckert Graduate Fellow	Spring 2026
Shell Outstanding Teaching Assistant Award	Spring 2026
NSF Graduate Research Fellowship Program - Honorable Mention	Spring 2025
2023 AIChE Undergraduate Research Symposium 1 st Place Awardee - Purdue University	Spring 2023
2022 Spring Undergraduate Research Conference 2 nd Place CoE Awardee - Purdue University	Spring 2022
First-Time Researcher Fellowship, Purdue University	Fall 2021
Purdue University Dean’s List and Semester Honors	2019 - 2023

Press

Georgia Tech: Study Uncovers New Insights into Nanoparticle Motion (06/05/2026)

Medical Press: Nanohydrogels steer cancer drugs to tumors, aiming to spare healthy tissue (03/10/2026)

Georgia Tech: Target the Tumor. Spare the Body. (03/06/2026)

Leadership, Involvement, and Service to Scientific Community

National Nanotech. Coordinated Infrastructure Research for Teachers Mentor Georgia Tech	Summer 2026
ChBE Graduate Student Recruitment Chair Georgia Tech	Fall 2024 - Spring 2026
Assistant Director TEDxPurdueU, Purdue University	April 2022 - April 2023
Chemical Engineering Ambassador Purdue University	April 2022 - May 2023
Peer Research Mentor Office of Undergraduate Research, Purdue University	Fall 2022

Additional Graduate Courses Taken

MSE 6110 (Transmission Electron Microscopy), MSE 8803 (Soft Nano and Biomaterials), PHYS 6205 (Soft Matter Physics), ChBE 8803 (AI for Chemical Engineers)

Technical Skills

Materials Characterization: Electron microscopy (TEM, SEM, EDS, in situ liquid-phase TEM), optical microscopy (brightfield, darkfield, fluorescence), Raman, XRD, XRF, DLS and zeta potential, UV-Vis, XPS, passive microrheology and nanorheology

Materials Synthesis & Processing: Nanoparticle synthesis and surface functionalization, solution-based thin film deposition, polymer coacervate formulation, active colloid fabrication

Data Analysis & Modeling: Python (NumPy, pandas, SciPy, trackpy), Github, deep learning (model fine-tuning, image segmentation), single-particle tracking, statistical diffusion analysis